

Judge Comments:

This is a very solid report that addresses all the elements to describe the team organization, project management, development approach, how the design was selected, how the design is expected to perform, the manufacturing and test plans and the test results demonstrating aircraft and subcomponent performance.

Some key positive highlights include:

- Overall very thorough coverage in each section
- Great to see plots of key variables from the performance prediction model in Figure 4.12, this really illustrates the fidelity of the model and gives some key parameters to validate against flight test data
- Good to consider weight/cg for each discrete condition of payload in Table 5.3, this highlights the team's thoroughness
- Good use of FEA to show structural integrity with factors of safety
- Good testing on the package deployment to determine the allowable ramp/drop conditions

Also noted a few minor areas of improvement needed:

- While it is clear that static stability is considered in the design, the static margin is not specified in the report
- No analysis given in Detailed Design for how vial packages will be deployed while staying within shock sensor threshold - Drawing package does not show how syringes will be stored/secured
- Drawing package systems layout drawing is missing some components such as speed controller, receiver, flight control servos - Should explain difference between flight test results and predictions (e.g. why were lap times were longer in flight test? Is this within the expected range of model uncertainty?)

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Be aware of significant digits. Many values are given to 4 or 5 significant digits. For example is the predicted score of 6.283 accurate to with 1/1000? Similar for 2.192 min (131.52 sec), or short period frequency of 13.9864 rad/s Management Summary: Well-written Just say no to 'kgf', it should be in the same dustbin as lb-mass. But you do get the prize for best airplane name. Conceptual Design: Good translation of mission requirements into design parameters Preliminary Design: Interesting that your final tail sizes were lower than original values based on typical tail volumes.

Judge Comments:

Nicely done! This report was well written and cohesive. Points were docked for the following:

- Executive Summary: no discussion of manufacturing plan and little discussion of testing plan.
- Management Summary: no listing of full team, just leads. "Avionics" used rather than "Propulsion and Electrical" makes me question understanding of the term. Overall Manufacturing schedule bar doesn't match sub-task detail schedules. No schedule information for sub-component design or test.
- Conceptual Design: No discussion of power/electric concept design.
- Preliminary Design: Multiple typos in this section. Too much reliance on past teams' designs as an explanation for why you chose a certain path. Lack of background on trade studies for control surface design. Multiple symbols and acronyms not defined (see section 4.4.6, ESC).
- Detail Design: Prior sections mention winglets, yet they are not shown in figures in this section. No discussion of how the fuselage lid will be secured. Description of propulsion and electrical system is

lacking. Description of control system is lacking. No mention of Ground Mission performance in the chart.

- Drawing Package: Winglets, control surface servos/hinges, fuselage lid attach, and syringe storage are not shown.